

Attention Is All You Need - Study Summary

Overview

This document is a study summary of the landmark 2017 paper "Attention Is All You Need" by Vaswani et al. It introduces the Transformer, a neural sequence model built entirely on attention mechanisms, removing recurrence entirely.

Motivation

Recurrent models such as LSTM and GRU process tokens one at a time, which prevents parallelization across sequence positions during training. For long sequences this becomes a serious bottleneck. The Transformer removes all recurrence and instead relies on a mechanism called self-attention, which lets every position attend to every other position in a single layer.

Architecture of the Transformer

Encoder Stack

The encoder is a stack of six identical layers. Each layer contains two sub-layers: a multi-head self-attention mechanism and a position-wise fully connected feed-forward network. Residual connections plus layer normalization are applied around each sub-layer.

Decoder Stack

The decoder also has six layers but adds a third sub-layer which performs multi-head attention over the output of the encoder stack. The decoder attends to previously generated tokens with masking to preserve autoregression.

Scaled Dot-Product Attention

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$. The scaling factor $\sqrt{d_k}$ counteracts the growth of dot products for large dimensions, which would otherwise push softmax into regions with tiny gradients.

Training Details and Results

Datasets

Experiments use two machine translation corpora: WMT 2014 English-German (4.5 million sentence pairs) and WMT 2014 English-French (36 million pairs). Sentences are encoded as byte-pair embeddings with a shared vocabulary of about 37000 tokens.

Hardware and Schedule

Models train on eight NVIDIA P100 GPUs. The base model trains for 12 hours (100000 steps) and the big model for 3.5 days (300000 steps).

Results

The base Transformer reaches 27.3 BLEU on English-to-German and 38.1 BLEU on English-to-French, beating all prior single models at a fraction of the training cost. The big model reaches 28.4 BLEU on English-German.